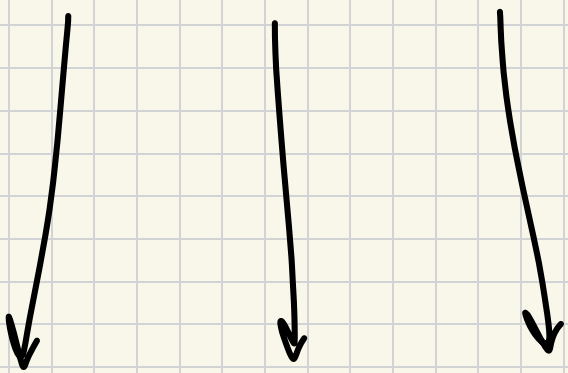
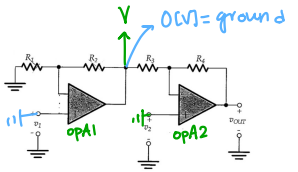
OP AMP PROBLEMS



1. (30 points) For the following difference amplifier, $R_1 = R_4 = 9\ k\Omega$ and $R_2 = R_3 = 1\ k\Omega$. Assume ideal op-amps. Find
   1. A_d,
   2. A_{cm} when $R_3 = 1.03\ k\Omega$,
   3. the CMRR, and
   4. the two input resistances (as seen at the two input terminals).

Hints: Use superposition. For calculating the CMRR, use the A_d from the first part – there is no need to attempt to recalculate it for the circuit with the new resistance. *Important:* show your work in a way that allows me to see how you solved this problem to receive credit.



$R_1 = R_4 = 9\ k\Omega$

$R_2 = R_3 = 1\ k\Omega$

$V_1 \neq 0$, $V_2 = 0$

opA1 → non-inverting amplifier

$V = V_1 \left(1 + \frac{R_2}{R_1}\right)$

opA2 → inverting amplifier

$V_{o_1} = V \cdot -\frac{R_4}{R_3} = -V_1 \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)$

$V_2 \neq 0$, $V_1 = 0$

opA1 → output is 0, ground

opA2 → non-inverting amplifier

$V_{o_2} = V_2 \left(1 + \frac{R_4}{R_3}\right)$

$V_{o,TOTAL} = V_{o_1} + V_{o_2}$

$V_{OUT} = -V_1 \underbrace{\frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)}_{\frac{9k\Omega}{1k\Omega}\left(1 + \frac{1k\Omega}{9k\Omega}\right) = 10} + V_2 \underbrace{\left(1 + \frac{R_4}{R_3}\right)}_{1 + \frac{9k\Omega}{1k\Omega} = 10}$

$V_{TOT} = (V_2 - V_1)\underbrace{10}_{A_d}$ $\quad A_d = 10$

Acm

$V_1 = V_2 = V_{cm}$

$V_{TOT} = V_2\left(1 + \frac{R_4}{R_3}\right) - V_1 \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)$

$V_{TOT} = V_{cm}\left(1 + \frac{R_4}{R_3}\right) - V_{cm} \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)$

$V_{TOT} = V_{cm}\left(\left(1 + \frac{R_4}{R_3}\right) - \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)\right)$

$A_{cm} = \frac{V_{TOT}}{V_{cm}} = \left(1 + \frac{R_4}{R_3}\right) - \left(\frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)\right)$

$= \left(1 + \frac{9k\Omega}{1.03k\Omega}\right) - \left(\frac{9k\Omega}{1.03k\Omega}\left(1 + \frac{1k\Omega}{9k\Omega}\right)\right)$

$A_{cm} = 0.029$

CMRR

$CMRR = 20\log\left(\frac{A_d}{A_{cm}}\right)$

$= 20\log\left(\frac{10}{0.029}\right)$

$CMRR = 50\ dB$

Input Resistances

- seen by input V_1, $R_{in,1} = \infty$
- seen by input V_2, $R_{in,2} = \infty$

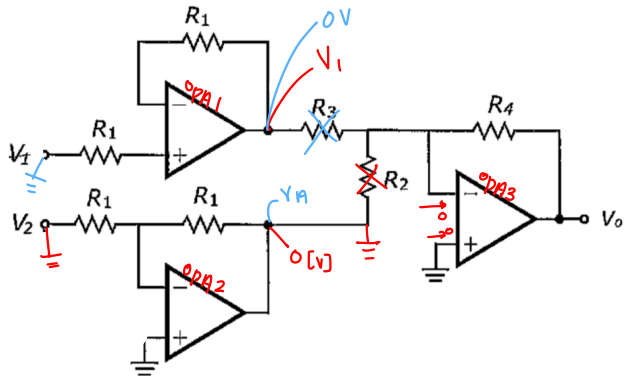
$1 + \frac{R_4}{R_3} = \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)$

$1 + \frac{R_4}{R_3} = \frac{R_4}{R_3} + \frac{R_2 R_4}{R_1 R_3}$

$1 = \underbrace{\frac{R_2 R_4}{R_1 R_3}}$

condition for the circuit to work as a difference amplifier

1. (35 points) For the circuit below



(a) What are the conditions for resistor values so that the circuit operates as a difference amplifier?

$V_1 \neq 0,\ V_2 = 0$

opA1 → voltage follower

output = V_1

opA3 → inverting amplifier

$V_{o1} = -V_1 \frac{R_4}{R_3}$

$V_1 = 0,\ V_2 \neq 0$

opA2 → inverting amplifier

$V_A = -V_2 \frac{R_1}{R_1} = -V_2$

opA3 → inv amp

$V_{o2} = -V_A \frac{R_4}{R_2}$

$V_{o2} = V_2 \frac{R_4}{R_2}$

$V_{TOT} = V_2 \frac{R_4}{R_2} - V_1 \frac{R_4}{R_3}$

for a difference amplifier

$\frac{R_4}{R_2} = \frac{R_4}{R_3}$

$R_2 = R_3$

(b) What is the expression for the respective difference gain, A_D?

$A_d = \frac{V_{TOT}}{V_d}$, $V_{TOT} =$

$V_{TOT} = V_2 A_d - V_1 A_d$

$V_{TOT} = (V_2 - V_1)\left(\frac{R_4}{R_2} - \frac{R_4}{R_3}\right)$

$V_{TOT} = V_d \left(\frac{R_4}{R_2} - \frac{R_4}{R_3}\right)$

$A_d = \frac{V_{TOT}}{V_d} = \frac{R_4}{R_2} = \frac{R_4}{R_3}$

(c) Common-mode gain, A_{CM}, of an ideal difference amplifier is equal to zero. What is the expression for the common-mode gain, A_{CM}, if conditions found in part (a) are not exactly satisfied?

$A_{cm} = \frac{V_{TOT}}{V_{cm}}$, $V_{cm} = V_1 = V_2$

$V_{TOT} = V_{cm} \cdot \frac{R_4}{R_2} - V_{cm} \frac{R_4}{R_3}$

$V_{TOT} = V_{cm}\left(\frac{R_4}{R_2} - \frac{R_4}{R_3}\right)$

$A_{cm} = \frac{R_4}{R_2} - \frac{R_4}{R_3}$

$A_{cm} = \frac{R_4}{R_2} - \frac{R_4}{R_3}$

(d) What are the input resistances seen at each input?

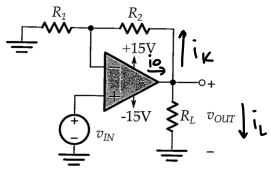
Input V_1 → since this opamp is just a voltage follower

$R_{in,1} = \infty$

Input V_2 → input resistance is just R_1

$R_{in,2} = R_1$

2. (40 points) Determine if the op-amp is in saturation (either current or voltage saturation) – you must show work that indicates that you have tested for each case. What is the voltage v_{OUT}, assuming an ideal op-amp? Here, $R_1 = 1\ k\Omega$, $R_2 = 9\ k\Omega$, $R_L = 2\ k\Omega$, $v_{IN} = 2$ V, and $i_{max} = 10$ mA.



1) assuming ideal

→ op amp is in the non-inverting configuration

$$V_{out} = V_{in}\left(1 + \frac{R_2}{R_1}\right)$$

$$V_{out} = V_{in}\left(1 + \frac{9k\Omega}{1k\Omega}\right)$$

$$V_{out} = 10\,V_{in}$$

---

2) given the bounds

$$-15\ V < V_{out} < +15\ V$$

$$i_{max} = 10\ [mA]$$

---

3) check voltage saturation when $V_{in} = 2\ [V]$

$$V_{out} = 10(2[V])$$

$V_{out} = 20[V] > 15[V]$ → voltage saturation

$V_{out} = 15[V]$

---

4) check current saturation when $V_{out} = 15[V]$

$$i_o = \frac{V_o}{R_2 + R_1} + \frac{V_o}{R_L}$$

when deriving this equation its important to note there is no Virtual Short between + & - terminals

$R_1 = 1k\Omega$, $R_2 = 9k\Omega$, $R_L = 2k\Omega$

$$i_o = \frac{15\ V}{9k\Omega + 1k\Omega} + \frac{15V}{2k\Omega}$$

$$i_o = 9mA < 10\ [mA]$$

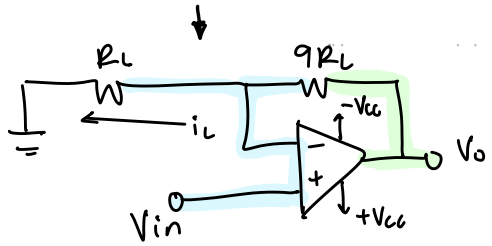
↖ no current saturation

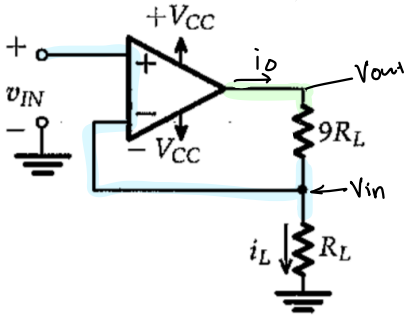
Voltage saturation

$V_{out} = 15[V]$

2. (35 points) In the circuit below, the op-amp is powered by a pair of $\pm V_{CC} = \pm 15$ Volt supplies. The maximum current for the op-amp is 15mA.

How we normally see it





odd configuration of a non inverting op-amp

$V_{out} = V_{in}\left(1 + \frac{9R_L}{R_L}\right)$

$V_{out} = V_{in} \cdot 10$

(a) Derive an expression for i_L in terms of v_{IN} and R_L.

$i_L = \frac{V_{in}}{R_L}$

(b) What is the range of v_{IN} that ensures the circuit does not go into voltage saturation?

$-15V < V_{out} < +15$

from ideal : $V_{out} = V_{in} \cdot 10$

$-15 < V_{in} \cdot 10 < +15$

$-1.5V < V_{in} < 1.5V$

(c) What is the range of load resistor values (R_L) that will ensure that the circuit does not go into current saturation for the full range of v_{IN} found in part (b)?

$\rightarrow i_{max} = 15[mA]$

$i_L = i_o = \frac{V_{in}}{R_L}$

$R_L = \frac{1.5V}{0.015mA}$

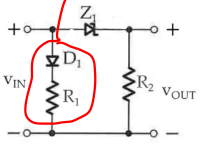
$R_L = 100\Omega$ ← this is the minimum value R_L needs to be in order to prevent current saturation

$R_L \geq 100[\Omega]$

DIODE PROBLEMS



3. (30 points) For the circuit below, assume $V_{D0} = 0.5$ volts, $R_1 = r_D$, $R_2 = 2r_Z$, $r_D = r_Z$, and $V_{Z0} = 2$ volts.



1. Write an expression for v_{OUT} in terms of v_{IN}.

doesn't contribute to Vout

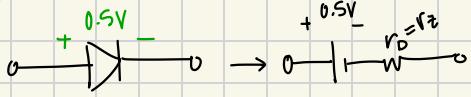
$R_1 = r_D$, $R_2 = 2r_Z$, $r_D = r_Z$

$V_{D0} = 0.5V$

$V_{Z0} = 2$ [V]

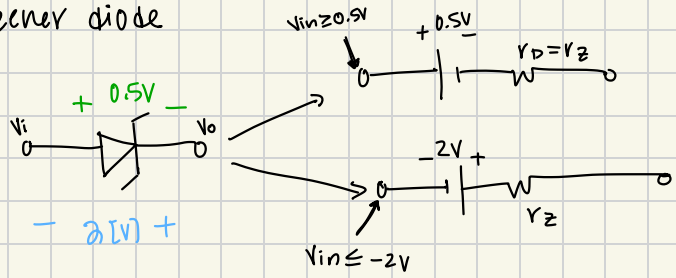
---

regular diode



---

zener diode



$V_i = V_p$

$V_o = V_n$

FB: $V_p - V_n \geq 0.5V$

$V_i - V_o \geq 0.5$ [V]

$V_i \geq 0.5$ [V]

RBB: $V_n - V_p \geq 2$ [V]

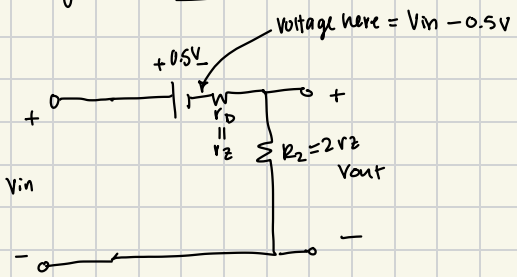
$V_o - V_i \geq 2$ [V]

$-V_i \geq 2V - V_o$

$V_i \leq -2$ [V]

---

1) zener is regular , $V_{in} \geq 0.5$ [V]



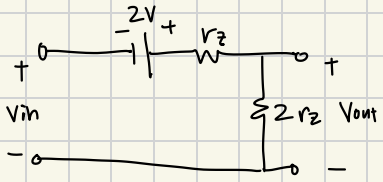
Voltage divider

$V_{out} = (V_{in} - 0.5[V]) \cdot \frac{2r_z}{3r_z}$

$V_{out} = \frac{2}{3}V_{in} - \frac{1}{3}$ // for $V_{in} \geq 0.5$ [V]

---

2) zener is reverse biasing , $V_{in} \leq -2$ [V]

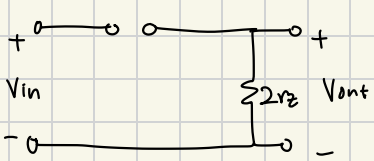


$V_{out} = (V_{in} + 2[V]) \cdot \frac{2r_z}{3r_z}$

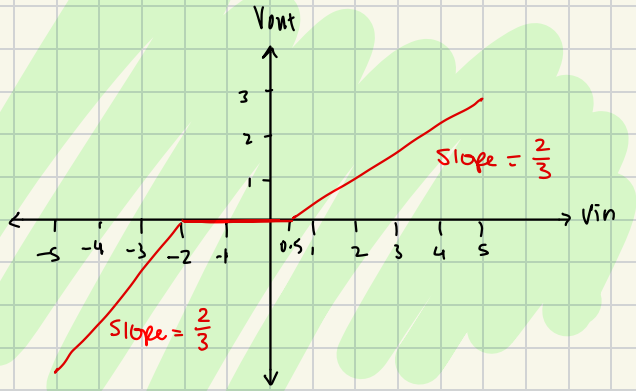
$V_{out} = \frac{2}{3}V_{in} + \frac{4}{3}$ // $V_{in} \leq -2$ [V]

---

3) Vin very small , $-2 < V_{in} < 0.5$

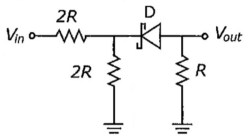


$V_{out} = 0$ // for $-2 < V_{in} < 0.5$



$$V_o = \begin{cases} \frac{2}{3}V_{in} + \frac{4}{3} & V_{in} \leq -2\,V \\ 0 & -2 < V_{in} < 0.5 \\ \frac{2}{3}V_{in} - \frac{1}{3} & V_{in} \geq 0.5 \end{cases}$$

3. (30 points) For the circuit below, assume the constant voltage drop model with $V_{D0} = 0.7\,V$, $V_{Z0} = 1.4\,V$, $r_D = r_z = R$



(a) Write an expression for v_{OUT} in terms of v_{IN}.

this (0.7 [V]) for the regular diode is the minimum required applied voltage for the diode to conduct

$V_{D0} = 0.7\,[V]$, $r_D = R$

$V_{Z0} = 1.4\,[V]$, $r_Z = R$

$V_p - V_n \geq 0.7$ } condition for forward bias mode

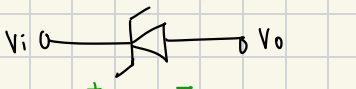
$V_o - V_i \geq 0.7$

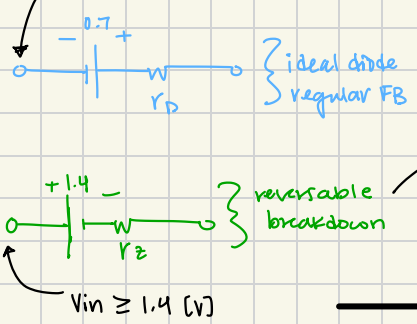
$-V_i \geq 0.7 - V_o$

$V_i \leq -0.7$

$V_{in} \leq -0.7\,[V]$

zener diode





min required voltage to force diode into reversable breakdown

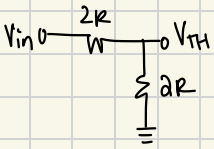
$V_n - V_p \geq 1.4$

$V_{in} - V_o \geq 1.4$

$V_{in} \geq 1.4$

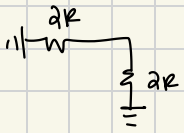
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1) find thevenin eq of the left side



$V_{TH} = V_{in}\frac{2R}{4R}$

$V_{TH} = V_{in}/2$



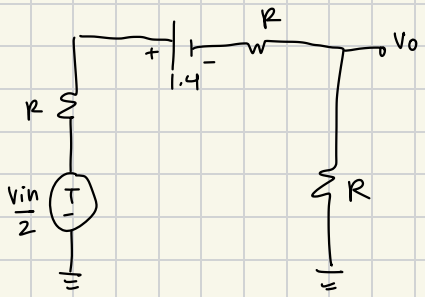
$R_{TH} = 2R // 2R = R$

---

2) assume $V_{in} \geq 1.4$

$V_{TH} = \frac{V_{in}}{2}$

$V_{in} \geq 2.8$



$V_o = \left(\frac{V_{in}}{2} - 1.4\right)\frac{R}{3R}$

$V_o = \frac{V_{in}}{6} - \frac{7}{15}\,[V]$ //

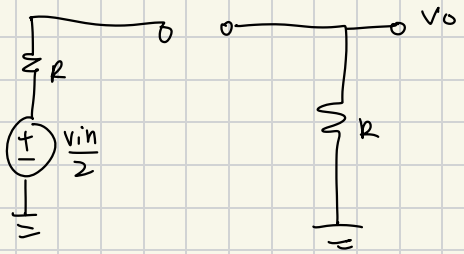
$V_{in} \geq 2.8\,[V]$

---

3) assume $-0.7 < V_{in} < 1.4$

$\downarrow$

$-1.4 < V_{in} < 2.8$ } because we used $\frac{V_{in}}{2}$ not just V_{in}



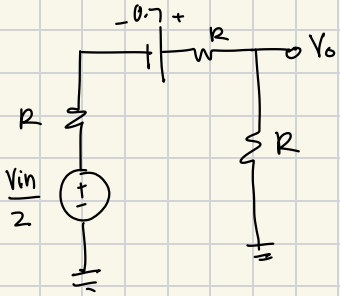
$V_o = 0$

$-1.4V < V_{in} < 2.8V$

---

a) assume $V_{in} \leq -0.7\,[V]$

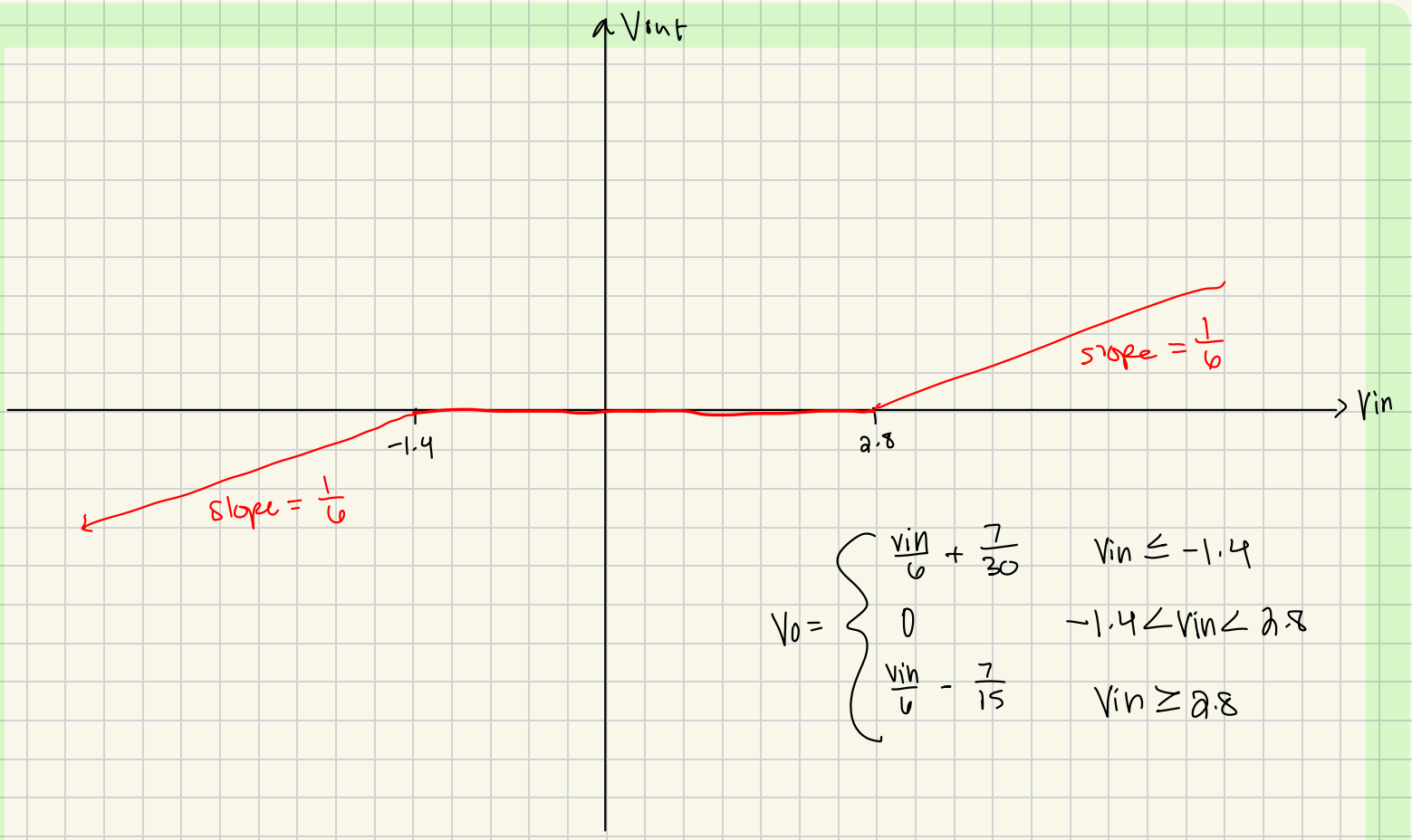
→ note since we made the thevenin equivalent $V_{TH} = V_{in}/2$ the bound becomes $V_{in} \leq -1.4\,[V]$



$V_o = \left(\frac{V_{in}}{2} + 0.7\right)\frac{R}{3R}$

$V_o = \frac{1}{6}V_{in} + \frac{7}{30}\,[V]$ //

$V_{in} \leq -1.4\,[V]$



$$V_o = \begin{cases} \frac{v_{in}}{6} + \frac{7}{30} & v_{in} \le -1.4 \\ 0 & -1.4 < v_{in} < 2.8 \\ \frac{v_{in}}{6} - \frac{7}{15} & v_{in} \ge 2.8 \end{cases}$$